

# Gerald meeting questions | 01-04

4 carriers / 8 ASICs / XC7A200T. Confirm the actual revision and requirements; leave unknown answers with a named measurement or source.

## **G01. Which exact ASIC and carrier revisions are current, and which files are authoritative?**

**Bring / answer:** Revision/date, released schematic, pin map, known changes from slides/repository.

**Why:** Avoid designing to an old revision.

**Owner / source:** Gerald / ASIC designer; versioned design archive.

Answer / file / next step: \_\_\_\_\_

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## **G02. Can we get the connector-to-ASIC contact table and exact mating connector/orientation?**

**Bring / answer:** Contact number, signal, ASIC identity, direction, power/ground role, part numbers and pin-1 drawing.

**Why:** Closes connector and routing correctness.

**Owner / source:** Gerald + carrier designer; CAD/pin sheet, then continuity check.

Answer / file / next step: \_\_\_\_\_

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## **G03. Is 1.5 V digital I/O correct for this ASIC revision, and what are its guaranteed limits?**

**Bring / answer:** Nominal/tolerance, VIH/VIL, VOH/VOL, absolute maximum, behavior while unpowered.

**Why:** Selects FPGA bank voltage and any level translation/isolation.

**Owner / source:** ASIC designer specification; Gerald identifies the source.

Answer / file / next step: \_\_\_\_\_

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## **G04. What clock-to-data, setup/hold, edge and duty-cycle limits apply at the carrier connector?**

**Bring / answer:** Timing diagram with minimum/maximum values and returned-clock relationship.

**Why:** Enables FPGA timing constraints and PCB skew budget.

**Owner / source:** ASIC specification/characterization; our team measures the implementation.

Answer / file / next step: \_\_\_\_\_

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# Gerald meeting questions | 05-08

**G05. Has this ASIC/carrier been validated at 32 MHz / 31.25 kS/s/channel, and why is reference code at 16 MHz?**

**Bring / answer:** Tested modes, measured sample rates and reason for lower-rate code.

**Why:** Separates target bandwidth from already-demonstrated silicon operation.

**Owner / source:** Gerald / hardware-test owner; test logs and reference firmware.

Answer / file / next step: \_\_\_\_\_

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**G06. What reset, clock-start and gain-programming sequence is required?**

**Bring / answer:** Polarities, pulse lengths, startup order, settling delay and known-working configuration.

**Why:** Determines boot logic and safe pull levels.

**Owner / source:** Gerald / ASIC designer; operating procedure plus bench verification.

Answer / file / next step: \_\_\_\_\_

\_\_\_\_\_

**G07. Which clocks/resets/programming/stimulation signals can be shared across eight ASICs?**

**Bring / answer:** Mark each shared/per-chip; fanout limit and inter-chip synchronization tolerance.

**Why:** Closes total signal count and clock buffering requirements.

**Owner / source:** Gerald for experiment; ASIC designer for electrical limits; we design buffers.

Answer / file / next step: \_\_\_\_\_

\_\_\_\_\_

**G08. Where are the left/right programming map, gain calibration and AC\_IN/IMP\_TEST pin specifications?**

**Bring / answer:** Bit/register tables, legal defaults, gain table, auxiliary pin role/range/unused termination.

**Why:** Prevents sending correctly timed but semantically wrong settings.

**Owner / source:** Gerald / ASIC designer; configuration files and pin specifications.

Answer / file / next step: \_\_\_\_\_

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# Gerald meeting questions | 09-12

**G09. Are five 1.5 V rails, VDD\_CGEN=1.6 V and REF\_VCM=0.75 V correct for the current chip?**

**Bring / answer:** Each rail name, approved voltage/tolerance and load served.

**Why:** Confirms the resistor-derived setpoints before reuse.

**Owner / source:** Gerald / ASIC designer; validated schematic and specification.

Answer / file / next step: \_\_\_\_\_

**G10. Do we have typical and peak current per rail in reset, recording and stimulation?**

**Bring / answer:** Per-ASIC table with clock, channels and operating condition; if absent, identify board/test points.

**Why:** Allows regulators, traces and thermal estimates to be sized.

**Owner / source:** Existing characterization/simulation, or measurements our team performs.

Answer / file / next step: \_\_\_\_\_

**G11. What noise/ripple, startup ramp/order and shutdown requirements apply?**

**Bring / answer:** Numerical noise limit and bandwidth; allowed rail differences, sequencing and ramp limits.

**Why:** Defines regulator/filter design and a measurable acceptance test.

**Owner / source:** ASIC designer requirements; our team verifies the power system.

Answer / file / next step: \_\_\_\_\_

**G12. Should VDD10 be +9 V or +10 V, and is R9 really 90 kohm on the tested board?**

**Bring / answer:** Approved VDD10 and INPUT\_VDD10 values, assembled R9 value and tested revision.

**Why:** The provided U8 SET calculation gives +9 V despite its name.

**Owner / source:** Gerald / regulator-board designer; actual measurement and released files.

Answer / file / next step: \_\_\_\_\_

# Gerald meeting questions | 13-16

## G13. What should INPUT\_VSS10 and VSS10 be, and is U9/R8 the validated topology?

**Bring / answer:** Input/output voltages, current direction, known-working schematic and measured output.

**Why:** The floating positive LDO does not simply create -10 V.

**Owner / source:** Gerald / regulator designer; LT3042 analysis and bench check.

Answer / file / next step: \_\_\_\_\_

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## G14. Which exact LT3042 package was assembled, and is there a corrected symbol-to-pad map?

**Bring / answer:** Complete part number, MSOP/DFN package, pin 1-10/EP mapping, released PCB/BOM.

**Why:** Old symbols use MSE numbering while footprint fields specify DFN.

**Owner / source:** Gerald / board designer; assembly records and ADI package pinouts.

Answer / file / next step: \_\_\_\_\_

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## G15. Which input feed is intended, and where are AGND/DGND/shields joined?

**Bring / answer:** Resolve old +2 V versus 3V3 input; give voltage tolerance and exact ground junction/location.

**Why:** Avoids combining incompatible references or incomplete return paths.

**Owner / source:** Gerald identifies working pairing; we check full schematic and continuity.

Answer / file / next step: \_\_\_\_\_

\_\_\_\_\_

## G16. Which rails must be local/per-chip, and which may share regulators?

**Bring / answer:** Allowed groups, supporting noise/crosstalk results and an agreed test if unknown.

**Why:** Determines whether 7, 28 or 56 low-voltage LDOs are needed and their overhead.

**Owner / source:** Gerald for signal-quality goals; ASIC designer and our measurements for sharing.

Answer / file / next step: \_\_\_\_\_

\_\_\_\_\_

# Gerald meeting questions | 17-20

## G17. What concurrent stimulation must the first board support?

**Bring / answer:** Active outputs per chip/stack, maximum current, pulse duration, repetition and duty cycle.

**Why:** Defines peak and average supply load; chip capability alone is insufficient.

**Owner / source:** Gerald / experiment owner; we calculate and test resulting power.

Answer / file / next step: \_\_\_\_\_

\_\_\_\_\_

## G18. Which configuration files and STIM\_EN/CLK/START/CHB waveforms are known to work?

**Bring / answer:** Example bitstreams/spreadsheets, pulse count and expected timing traces.

**Why:** Verifies actual control semantics rather than relying on comments.

**Owner / source:** Gerald / original control author; our FPGA implementation reproduces it.

Answer / file / next step: \_\_\_\_\_

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## G19. What samples are invalid during/after stimulation, and how long until recovery?

**Bring / answer:** Reset/blanking timing, valid-data delay and required metadata marking.

**Why:** A continuous USB stream does not mean continuously valid neural data.

**Owner / source:** Gerald / ASIC characterization; our firmware marks and verifies data.

Answer / file / next step: \_\_\_\_\_

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## G20. What states are required at power-up, lost configuration, stopped clock, cable disconnect and reset?

**Bring / answer:** Control levels, electrode/stimulation state and charge-balance requirements.

**Why:** Determines pull resistors and interlocks; validate on dummy loads.

**Owner / source:** Gerald / ASIC designer operating procedure; our team tests behavior.

Answer / file / next step: \_\_\_\_\_

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# Gerald meeting questions | 21-24

## G21. Which carrier dimensions and keep-outs must remain despite flexible board size?

**Bring / answer:** Connector position/height, wire-bond clearance, thickness, mounts and flex bend limits.

**Why:** Keeps the new board compatible with the existing stack.

**Owner / source:** Carrier designer/Zitong + Gerald; CAD and measured hardware.

Answer / file / next step: \_\_\_\_\_

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## G22. How long must a recording run, and what should happen if a downstream stall fills the buffer?

**Bring / answer:** Required run duration; permitted losses if any; stop/flag/drop behavior and maximum tolerated gap.

**Why:** Determines RAM capacity, error handling and storage requirements.

**Owner / source:** Gerald/experiment owner defines behavior; our team measures actual stalls.

Answer / file / next step: \_\_\_\_\_

\_\_\_\_\_

## G23. How accurately must all ASICs and any external experiment trigger be aligned?

**Bring / answer:** Allowed inter-chip/time-stamp error, trigger voltage/direction and event timing requirements.

**Why:** Determines clock/trigger connectors, synchronization logic and timestamp format.

**Owner / source:** Gerald/experiment owner; ASIC timing sources and our implementation tests.

Answer / file / next step: \_\_\_\_\_

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## G24. What measurable results define a successful eight-ASIC bench demonstration?

**Bring / answer:** Full-rate duration, allowed frame errors, gain/channel/noise checks and required stimulation tests.

**Why:** Turns completion into a repeatable test rather than a subjective impression.

**Owner / source:** Gerald agrees experiment criteria; our team supplies test results.

Answer / file / next step: \_\_\_\_\_

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# Checks our team owns

Gerald provides ASIC and experiment facts. These tasks are engineering work we can perform directly; they do not require asking him to guarantee vendor performance.

| Work                     | Deliverable                                                                                                                               |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| USB/FX3/KR260 path       | Measured $\geq 264$ MB/s for the current format, preferably $\geq 316.8$ MB/s with chosen margin, no sequence/CRC errors, bounded stalls. |
| FPGA pins and timing     | Legal package/bank/clock/DDR allocation, complete constraints and timing closure.                                                         |
| Power and thermal design | XPE/Vivado estimate plus measurements; exact regulators/support values and startup/loop/thermal checks.                                   |
| BOM completeness         | Schematic-derived populated quantities, exact order codes, matched footprints, no duplicate capacitor accounting.                         |
| PCB and assembly         | Stackup/escape/impedance and manufacturer constraints; ERC/DRC, connector orientation and assembly review.                                |

Priority evidence to bring back: current pin map; ASIC voltage/timing limits; measured rail currents; corrected high-voltage/package reference; regulator-sharing/noise requirements; stimulation and recording acceptance criteria.

Prepared 18 September 2026. The companion FPGA System Design and Parts Guide includes the original circuit excerpt and manufacturer source links.